

vax_11_780_cpu_sop_v1.json

```
{
  "id": "SOP-VAX-001",
  "title": "VAX-11/780 CPU Machine Check (MCHECK) Diagnostic",
  "system": "VAX-11/780",
  "component": "KA780 Central Processing Unit",
  "author": "SymbioStreams Semantic Bridge",
  "source": "DEC VAX Maintenance Handbook (EK-VAXV2-HB-002)",
  "description": "Diagnostic logic for identifying the cause of CPU Machine Checks (MCHECK) on the VAX-11/780 platform.",
  "steps": [
    {
      "step": 1,
      "action": "Machine Check Stack Frame Retrieval",
      "details": "When an MCHECK occurs, the hardware pushes a summary frame onto the stack. Extract the 'Summary Parameter' byte.",
      "indicators": {
        "Code 01": "CP Read Timeout (Control Store/Memory)",
        "Code 02": "CP Read Data Parity Error",
        "Code 05": "IB (Instruction Buffer) Parity Error"
      }
    },
    {
      "step": 2,
      "action": "Examine Internal CPU Registers",
      "details": "Use the LSI-11 console to 'EXAMINE' the Internal Processor Registers (IPR). Check IPR 32 (IDBUS) and IPR 35 (EALU).",
      "check": "Verify if the parity error is localized to the IDBUS or the ALU output."
    },
    {
      "step": 3,
      "action": "Control Store Diagnostic",
      "details": "If the error code is 01, the microcode path may be corrupt. Load the CPU Micro-diagnostics (EVKAA series) from the console floppy.",
      "logic": "Run the 'Control Store Checksum' test to verify the integrity of the WCS (Writable Control Store)."
    },
    {
      "step": 4,
      "action": "Module Level Isolation",
      "details": "The EVKAA series will provide a 3-digit error code. Cross-reference this with the Module Utilization Chart.",
      "remedy": "Isolate the failure to the specific CPU board (e.g., M8218 for Control Store or M8219 for ALU) and replace."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Summary Code 01",
      "next": "Check control store integrity and memory timing."
    },
    {
      "condition": "IB Parity Error".
    }
  ]
}
```

```
    "next": "Replace Instruction Buffer module (M8223)."  
  },  
  {  
    "condition": "Intermittent MCHECK",  
    "next": "Check CPU power supply voltage margins (+5V +/- 5%)."  
  }  
]  
}
```